

Total no. of Pages: 2

Roll no.

Degree: B.Tech (ECAM) Semester: III rd. - Course work
END-SEMESTER EXAMINATION, Nov-Dec 2023

Course Title: EAEPC06

Course Code: Probability theory and random process

Time: 03 Hours

Max. Marks: 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Describe uniform distribution. Draw the PDF and CDF of a uniformly distributed random variable. Find its mean and variance.	5	CO1
1b	Let X be a random variable with distribution $f(x) = \begin{cases} C + \frac{x}{6} & \text{if } 0 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$ i. Find the value of C. ii. Find the probability $P(1 \leq X \leq 2)$.	5	CO1
1c	A fair coin is tossed 2000 times. Find the probability that head will show 500 times.	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Explain joint distribution and joint density function with example.	5	CO2
2b	Define ij -th central moment of two random variables, X and Y . Explain the covariance of X and Y along with its significance. What is correlation coefficient?	5	CO2
2c	Explain marginal distribution, conditional distribution, and conditional expectation.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	X is a random variable with the PDF $f_X(x)$, and $Y = g(X)$. Derive the PDF of Y .	5	CO3
3b	The PDF of a random variable X is given by $f_X(x) = 0.5$, for $0 < x \leq 2$, and zero elsewhere. Determine the PDF of Y where $Y = 2X + 3$.	5	CO3
3c	Prove Chebyshev inequality.	5	CO3
Q 4	Attempt any 2 parts of the following		
4a	Explain Wide Sense Stationary (WSS) process and its properties.	5	CO4
4b	What is autocorrelation function of random process? Explain the properties of autocorrelation function.	5	CO4

4c	The autocorrelation function of a random process is given by, $R_{xx}(\tau) = 100 + \frac{9}{1+\tau^2}$. Find mean value and variance of the process	5	CO4
Q 5	Attempt any 2 parts of the following		
5a	Derive the relationship between the power density spectrum and the autocorrelation function of a signal.	5	CO5
5b	A WSS random process is applied as input to an LTI system. Prove that the output will be a WSS random process.	5	CO5
5c	Explain "white noise" along with its power density spectrum and autocorrelation function. Why is "white noise" unrealizable?	5	CO5

3rd Sem.-B.Tech.

END-SEMESTER EXAMINATION, NOV.-DEC.,2023

Course Code- EAMTC04

Course Title- Mathematics for Machine Learning

Time- 03 Hours

Max. Marks-50

Note: - Attempt all five questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer.

Q.No.	Attempt any 2 parts of the questions below	Marks	CO
1(a)	Find KerT & ImT for the linear mapping $T: R^2 \rightarrow R^3$ defined by $T(x, y) = (x + y, x - y, y)$.	05	CO1
1(b)	Show that $T: R^4 \rightarrow R^4$ $T(x, y, z, t) = (2x, 3y, 0, 0)$ is a linear transformation. Find its rank and nullity.	05	CO1
1(c)	Find a basis and its dimension for the subspace of R^3 spanned by the set $S = \{(3, 2, 1), (1, 2, 0), (-1, 2, -1)\}$	05	CO1
2(a)	Use the Gram-Schmidt process to convert the basis $\{(1, 0, 1), (1, 0, -1), (0, 3, 4)\}$ of the Euclidean space R^3 with standard inner product into an orthonormal basis.	05	CO2
2(b)	Find the eigen values, eigen vectors and eigen spaces of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	05	CO2
2(c)	Reduce the quadratic form $5x^2 + y^2 + 10z^2 - 4yz - 10zx$ to the normal form and show that it is positive definite.	05	CO2
3(a)	Find the polar decomposition of $A = \begin{pmatrix} 11 & -5 \\ -2 & 10 \end{pmatrix}$	05	CO3
3(b)	Find the singular value decomposition of the matrix $A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$	05	CO3
3(c)	Find Pseudo inverse of the matrix $A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{pmatrix}$	05	CO3
4(a)	Verify Green's Lemma in the xy plane for $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the closed curve of the region bounded by $y = \sqrt{x}, y = x^2$.	05	CO4
4(b)	Find $\text{div} \vec{F}$ and $\text{curl} \vec{F}$ where $\vec{F} = \text{grad} (x^3 + y^3 + z^3 - 3xyz)$	05	CO4
4(c)	If $\vec{F} = x\hat{i} + (z^2 - zx)\hat{j} - xy\hat{k}$. Evaluate $\iint_S \vec{F} \cdot \vec{ds}$ where S is the surface of the triangle with vertices (2,0,0), (0,2,0), (0,0,4)	05	CO4
5(a)	Examine if the set $S = \{(x, y) \in R^2: x^2 + y^2 \leq 4\}$ is a convex set.	05	CO5
5(b)	Minimize x Subject to $x^2 + y^2 = 1$	05	CO5
5(c)	Solve: Min $f(x) = -2x_1 - x_2$ subject to $x_1 - x_2 \leq 0$ $x_1^2 + x_2^2 \leq 4$ $x_1, x_2 \geq 0$	05	CO5

Total no. of Pages:

Roll no.

Degree: B. Tech Semester: Third Sem Course work
END-SEMESTER EXAMINATION, Nov-Dec 2023
 Course Title: Digital Circuits and Systems
 Course Code: EAEP08

Time: 03 Hours

Max. Marks: 40

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Explain the need for and importance of error detecting and error correcting code. Find the position of error in the following received data using even parity error detection assuming four- bit original data: Data received = (1110101) ₂	4	01
1b	Implement the following Boolean function F, together with the don't-care conditions d, using no more than two NOR gates. Assume that both the normal and complement inputs are available. F (A, B, C, D) = $\sum (2, 4, 10, 12, 14)$ D (A, B, C, D) = $\sum (10, 1, 5, 8)$	4	01
1c	i. Define the following parameters: a, Propagation Delay b, Fan-Out c, V _{IH} d, Rise time. ii. Draw the CMOS circuit for the given expression: F = ((A+B) C + A' B')'	2 2	02
Q 2	Attempt any 2 parts of the following		
2a	Design a 4-bit incrementor circuit	4	03
2b	Explain the need of carry-look ahead generator and draw a 4-bit carry-look ahead generator.	4	03
2c	i. Implement the given two functions using PLA with minimum number of product terms: F1(A, B, C) = $\sum m (0,1,2,4)$ F2(A, B, C) = $\sum m (0,5,6,7)$ ii. The inputs of a 4 to 1 MUX are as follows: I ₀ = Z, I ₁ = Z', I ₂ = '0', I ₃ = '1'. The select lines are connected to the inputs of the function X & Y. Z is also the input of the function. Find the function which the multiplexer implements	2 2	05 03
Q 3	Attempt any 2 parts of the following		
3a	What is the difference between ring counter and Johnson's ring counter. Draw the block diagram and timing diagram of 4-bit Johnson's ring counter and explain.	4	03

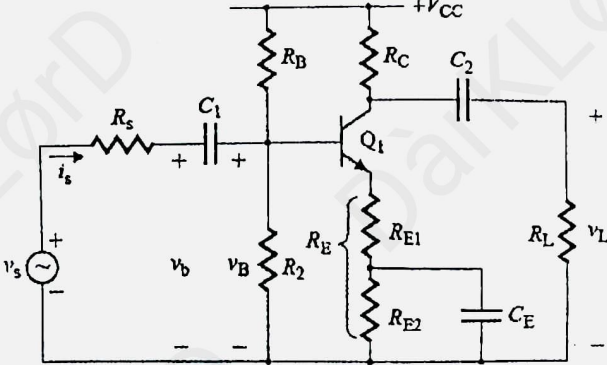
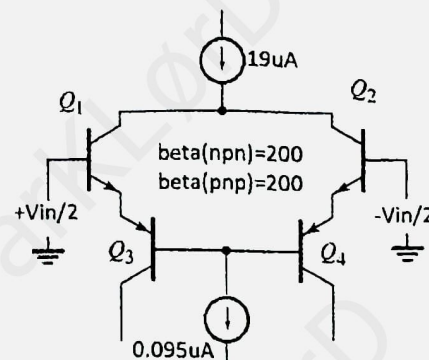
3b	Draw the Mod-10 ripple counter and explain its working with the help of timing diagram.	4	03
3c	Design a MOD-3 up-down synchronous counter using T- Flipflop.	4	05
Q 4	Attempt any 2 parts of the following		
4a	Explain the importance of digital circuit simulation using VHDL and explain the hardware abstraction using entity-architecture pair.	4	04
4b	Write the VHDL code of half adder using behavioural and structural modelling.	4	04
4c	Write the VHDL code for 4-bit SISO shift register.	4	04
Q 5	Attempt any 2 parts of the following		
5a	Design a serial adder using JK Flipflop.	4	05
5b	Draw the state diagram to detect the sequence "1101" using mealy model and implement using D-Flipflop.	4	05
5c	Write notes on the following: i.CPLD ii.FPGA	2 2	05

Degree: BTech (ECAM) Semester: IIIrd
END-SEMESTER EXAMINATION, Nov-Dec 2023
Course Title: Microelectronic Circuits and Applications
Course Code: EAEPC-07

Time: 03 Hours

Max. Marks: 40

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer. Make your explanations bulleted as far as possible.

Q. No.	Questions	Marks	CO
Q 1.	Attempt any 2 parts of the following		
1a	What are different modes of BJT's operation. Discuss those operation regions with a neat o/p characteristic of an npn transistor. Why V_{CEsat} is about 0.2V in a silicon npn BJT?	4	CO1
1b	Discuss with necessary equation the Triode and saturation regions of the MOSFET and also elaborate with necessary equation and neat diagram (including o/p characteristic) the channel length modulation.	4	CO1
1c	From the circuit diagram of Fig.1c calculate the Q-point values (I_{CQ} , V_{CEQ} , I_{BQ}) and input impedance and voltage gain. (Assume $V_{BEac1}=0.7V$ and $\beta=175$, $R_C=500\Omega$, $R_E=495\Omega$, $R_B=21K\Omega$, $R_2=8.5K\Omega$, $R_{E1}=22.2\Omega$, $R_{E2}=472.8\Omega$, all Cs are 100uF).	4	CO1
Q 2.	Attempt any 2 parts of the following		
2a	For the differential pair shown in Fig2a draw its small signal equivalent and compute the input impedance.	4	CO2
 Fig 1c  Fig 2a			
2b	For a BJT differential pair derive the hyperbolic relationship between the output differential current and input differential voltage. Also, derive compute its input impedance.	4	CO2
2c	Draw a neat circuit schematic of the KHN biquad and derive various second-order (low-pass, high-pass and band-pass) transfer functions.	4	CO2
Q 3.	Attempt any 2 parts of the following		
3a	Show that the circuit shown in Fig.3a is a non-inverting differentiator. Derive the expression for V_o/V_i .	4	CO3
3b	Derive an expression for $Z_{in}=(V_{in}/I_{in})$ for the circuit given in Fig.3b and draw the equivalent circuit of Z_{in} .	4	CO3
3c	Derive an expression for the circuit shown in Fig.3c. What approximation is needed to realize first-order all pass filter?	4	CO3
Q 4.	Attempt any 2 parts of the following		
4a	Draw the two types opamp-based Schmitt trigger circuits and discuss analytically these two in detail with necessary V_o versus V_i diagrams.	4	CO4

4b	Draw the opamp-based <i>triangular-cum-square</i> wave generator and derive the expression for the frequency of the wave form being generated.	4	CO4
4c	Discuss the limitations of linear applications of the bipolar OTA. Derive the expression for I_0 from the circuit of Fig.4c and comment on this result. (2-quadrant multiplier)	4	CO4
Q 5.	Attempt any 2 parts of the following		
5a	Discuss the types of feedback for <i>voltage, current, transconductance</i> and <i>transimpedance</i> amplifiers. Also enumerate with justification the merits of negative feedback.	4	CO5
5b	From the following block diagrams (Fig.5b1, Fig.5b2) check if they are oscillators, if yes, find out <i>condition of oscillation</i> and then <i>frequency of oscillation</i> .	4	CO5
5c	With an exemplary circuit diagram of Class-A amplifier discuss its working and the two important figure of merits (i) <i>efficiency</i> , (ii) <i>Harmonic distortion</i> in comparison to Class-B amplifier.	4	CO5

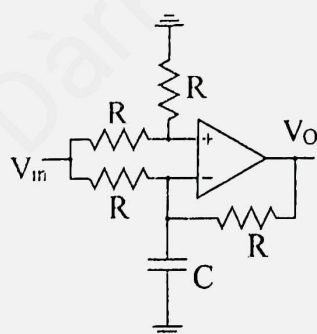


Fig.3a

(assume ideal opamps)

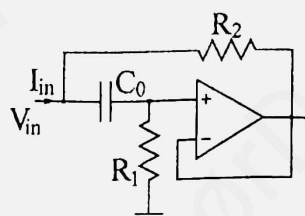


Fig.3b

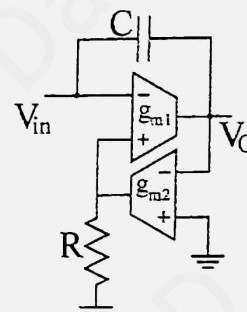


Fig.3c

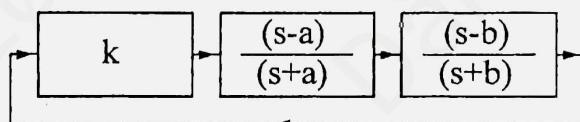


Fig.5b1

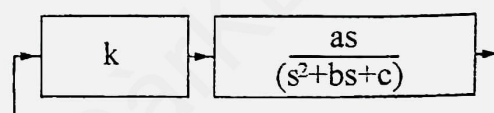


Fig.5b2

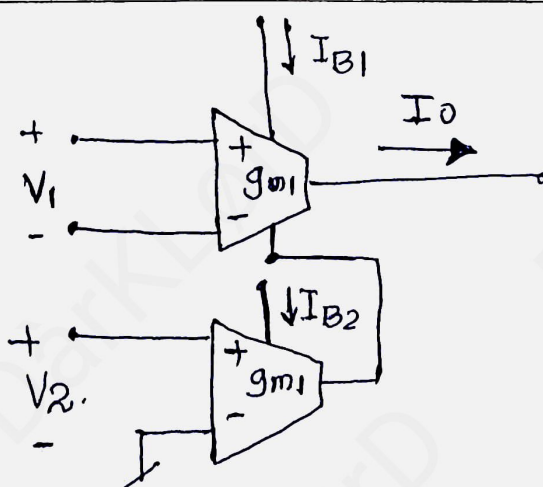


Fig.4c

III rd Sem.-B.Tech. END-SEMESTER EXAMINATION, NOV.-DEC.,2023 Course

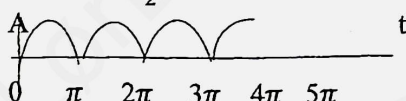
Code- EAEPC 05

Course Title- Signals and Systems

Time- 3 Hours

Max. Marks- 50

Note: Attempt all five questions. Missing data / information if any, may be suitably assumed & mentioned in the answer

Que	Attempt any two part of Questions BELOW	Marks	COs
1-a	Determine the power and Energy of the following signals: i) $x(t) = e^{-4t} u(t)$ ii) $y(n) = (\frac{1}{4})^n u(n)$ iii) $z(t) = \cos(t)$	5	CO 1
1-b	Determine if each of the following systems is invertible. If it is, construct the inverse system. If it is not, find two input signals to the system that have the same output. $y(t) = x(t-5)$ ii) $y[n] = \cos[x[n]]$	5	CO1
1-c	Explain advantages and disadvantages of discrete time systems over continuous time systems.	5	CO1
2-a	Input-output relationship for a system is given by $y[n] = x[n-2]x[n-4]$, determine whether the system is memoryless, invertible and linear.	5	CO2
2-b	A causal LTI system is characterized by the following difference Equation: $y[n] - (3/4)y[n-1] + (1/8)y[n-2] = 2x[n]$ Find out the impulse response.	5	CO2
2-c	Explain the properties of LTI system.	5	CO2
3-a	Find Fourier series of the following sequence- $f(t) = A \sin(\frac{\omega t}{2})$ 	5	CO3
3-b	Consider the rectangular pulse signal $X(t) = \begin{cases} 1 & t < T_1 \\ 0 & t > T_1 \end{cases}$ Find out its Fourier transform	5	CO3
3-c	Explain initial and final value theorem.	5	CO3
4-a	Consider a stable LTI system characterized by the differential equation- $\frac{dy(t)}{dt} + a y(t) = x(t)$, where $a > 0$; find out impulse response of the filter $h(t)$	5	CO4
4-b	Solve the differential equation given below using Laplace transform $\frac{d^2 t}{dt^2} + 4 \frac{dt}{dt} + 5i = 5 u(t)$ if given initial conditions are $i(0) = 1$ and $\frac{di(0)}{dt} = 2$	5	CO4
4-c	Determine $x(t)$ for the following conditions: a) $x(t)$ is right-sided b) $x(t)$ is left-sided $X(s) = \frac{1}{(s+1)(s+2)}$	5	CO4
5-a	Find the inverse Z transform of the following signal $X(Z) = \log(1+z/a)$	5	CO5
5-b	Find the z transform as well as ROC for the following sequence (i) $(1/3)^n u(-n)$ and (ii) $(1/3)^n [u(-n) - u(n-8)]$	5	CO5
5-c	Explain Parseval's relation for discrete time periodic signals.	5	CO5